



Vector Space

Department of Computer Engineering
Sharif University of Technology

Hamid R. Rabiee rabiee@sharif.edu

Maryam Ramezani maryam.ramezani@sharif.edu



Table of contents

01

Review

02

Field

03

Vector Space

04

Linear Combination

05

Span-Linear Hull



01

Review



Complex Numbers

Sum and Products

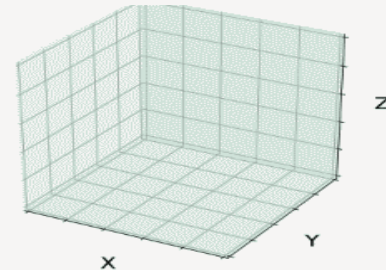
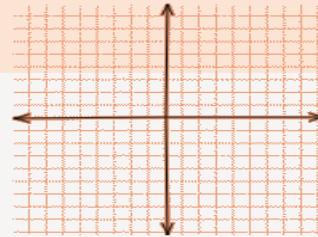
Tuple and Vector Space

Definition

- A tuple is an ordered list of numbers.
- For example: $\begin{bmatrix} 1 \\ 2 \\ 32 \\ 10 \end{bmatrix}$ is a 4-tuple (a tuple with 4 elements).

$$\mathbb{R}^2 = \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 0.112 \\ \frac{2}{3} \end{pmatrix}, \begin{pmatrix} \pi \\ e \end{pmatrix}, \dots \right\}$$

$$\mathbb{R}^3 = \left\{ \begin{pmatrix} 17 \\ \pi \\ 2 \end{pmatrix}, \begin{pmatrix} 9 \\ -2 \\ \sqrt{2} \end{pmatrix}, \begin{pmatrix} 1 \\ 22 \\ 2 \end{pmatrix}, \dots \right\}$$



Review: Complex Numbers

Numbers:

- Real: Nearly any number you can think of is a Real Number!

1	12.38	-0.8625	$3/4$	$\sqrt{2}$	1998
---	-------	---------	-------	------------	------

- Imaginary: When squared give a negative result.

The “unit” imaginary number (like 1 for Real Numbers) is “ i ”, which is the square root of -1 .

Examples of Imaginary Numbers:

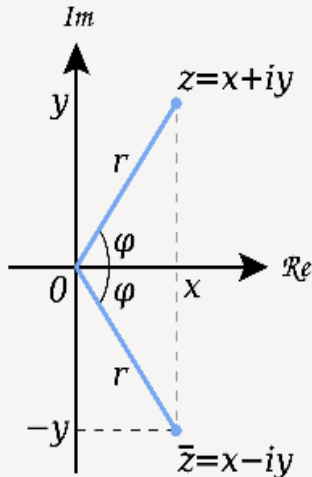
$3i$	$1.04i$	$-2.8i$	$3i/4$	$(\sqrt{2})i$	$1998i$
------	---------	---------	--------	---------------	---------

And we keep that little “ i ” there to remind us we need to multiply by $\sqrt{-1}$

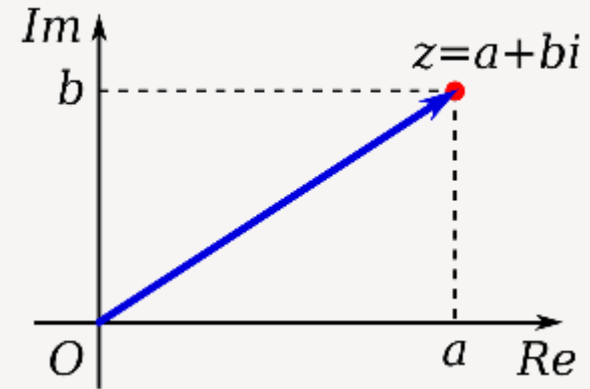
Review: Complex Numbers

- $\mathbb{C}$ is a plane, where number $(a + bi)$ has coordinates $\begin{bmatrix} a \\ b \end{bmatrix}$
- Imaginary number: $bi, b \in \mathbb{R}$

- □ Conjugate of $x + yi$ is noted by $\overline{x + yi}$:
 - $x - yi$



(Complex conjugate)



Review: Complex Numbers

□ Arithmetic with complex numbers $(a + bi)$:

□ $(a + bi) + (c + di) = (a + c) + (b + d)i$

□ $(a + bi)(c + di) = ac - bd + (bc + ad)i$

□ $\frac{a+bi}{c+di} = \frac{(a+bi)(c-di)}{(c+di)(c-di)} = \frac{ac+bd}{c^2+d^2} + \left(\frac{bc-ad}{c^2+d^2}\right)i$

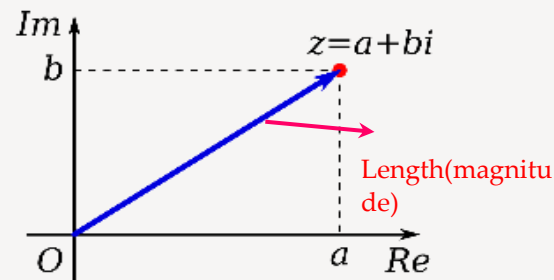
Review: Complex Numbers

□ Length (magnitude): $||a + bi||^2 = \overline{(a + bi)}(a + bi) = a^2 + b^2$

• Inner Product:

□ Real: $\langle x, y \rangle = x_1y_1 + x_2y_2 + \dots + x_ny_n$

□ Complex: $\langle x, y \rangle = \overline{x_1}y_1 + \overline{x_2}y_2 + \dots + \overline{x_n}y_n$



Extra resource:

If you want to learn more about complex numbers, this video is recommended!

Binary Operations

What is a binary operation?

Binary Operations

Definition

Any function from $A \times A \rightarrow A$ is a binary operation.



□ Closure Law:

- A set is said to be closure under an operation (like addition, subtraction, multiplication, etc.) if that operation is performed on elements of that set and result also lies in set.

$$\text{if } a \in A, b \in A \rightarrow a * b \in A$$



Binary Operations

Example

- ☐ Is “+” a binary operator on natural numbers?
- ☐ Is “ $\times$ ” a binary operator on natural numbers?
- ☐ Is “-” a binary operator on natural numbers?
- ☐ Is “/” a binary operator on natural numbers?

02

Field



Fields

$$\forall a, b, c \in F$$

Properties	Binary Operations	
	Addition (+)	Multiplication (.)
Closure (بسته بودن)	$\exists a + b \in F$	$\exists a.b \in F$
Associative (شرکت پذیری)	$a + (b + c) = (a + b) + c$	$a.(b.c) = (a.b).c$
Commutative (جابہ جایی پذیری)	$a + b = b + a$	$a.b = b.a$
Existence of identity $e, e' \in F$	$a + e = a = e + a$	$a.e' = a = e'.a$
Existence of inverse: For each a in F there <u>must exist</u> b in F	$a + b = e = b + a$	$a.b = e' = b.a$ <u>For any nonzero a</u>
Multiplication is distributive over addition $a.(b + c) = a.b + a.c$ $(a + b).c = a.c + b.c$		

Fields

- A field in mathematics is a set of things or elements (not necessarily numbers) for which the basic arithmetic operations (addition, subtraction, multiplication, division) are defined: $(F, +, \cdot)$



Example

$(\mathbb{R}; +, \cdot)$ and $(\mathbb{Q}; +, \cdot)$ serve as examples of fields.

- Field is a set (F) with two binary operations $(+, \cdot)$ satisfying following properties:
- 
- 

Fields

Example

Set $B = \{0,1\}$ under following operations is a field?



+	0	1
0	0	1
1	1	0

.	0	1
0	0	0
1	0	1

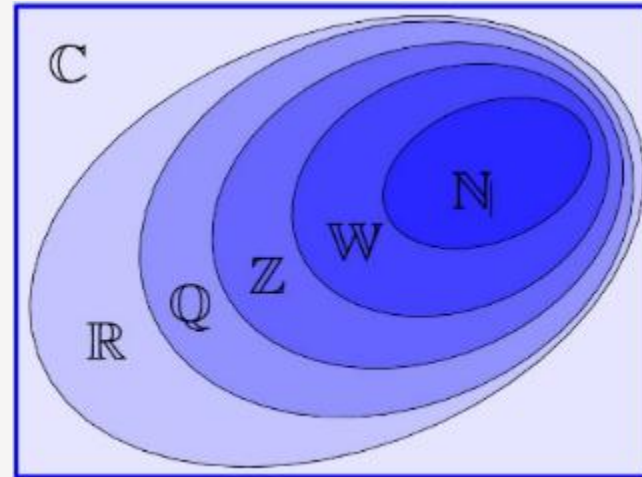


Fields

Example

Which are fields? (two binary operations $+$, $*$)

$\mathbb{R}$
 $\mathbb{C}$
 $\mathbb{Q}$
 $\mathbb{Z}$
 $\mathbb{W}$
 $\mathbb{N}$
 $\mathbb{R}^{2 \times 2}$



$\mathbb{C}$: Complex
 $\mathbb{R}$: Real
 $\mathbb{Q}$: Rational
 $\mathbb{Z}$: Integer
 $\mathbb{W}$: Whole
 $\mathbb{N}$: Natural

03

Vector Space



Vector Space

- Building blocks of linear algebra.
- A non-empty set V with field F (most of time $\mathbf{R}$ or $\mathbf{C}$) forms a vector space with two operations:
 1. $+$: Binary operation on V which is $V \times V \rightarrow V$
 2. $\cdot$: $F \times V \rightarrow V$

Note

In our course, by **default**, field is $\mathbf{R}$ (real numbers).

Vector Space

Definition

A vector space over a field F is the set V equipped with two operations: $(V, F, +, \cdot)$

- i. **Vector addition:** denoted by “+” adds two elements $x, y \in V$ to produce another element $x + y \in V$
- ii. **Scalar multiplication:** denoted by “ $\cdot$ ” multiplies a vector $x \in V$ with a scalar $\alpha \in F$ to produce another vector $\alpha \cdot x \in V$. We usually omit the “ $\cdot$ ” and simply write this vector as αx .

Vector Space Properties

□ Addition of vector space ($x + y$)

□ **Commutative** $x + y = y + x \quad \forall x, y \in V$

□ **Associative** $(x + y) + z = x + (y + z) \quad \forall x, y, z \in V$

□ **Additive identity** $\exists \mathbf{0} \in V$ such that $x + \mathbf{0} = x, \forall x \in V$

□ **Additive inverse** $\exists (-x) \in V$ such that $x + (-x) = \mathbf{0}, \forall x \in V$

Vector Space Properties

□ Action of the scalars field on the vector space

(αx)

□ **Associative** $\alpha(\beta x) = (\alpha\beta)x$ $\forall \alpha, \beta \in F; \forall x \in V$

□ **Distributive over**

scalar addition: $(\alpha + \beta)x = \alpha x + \beta x$ $\forall \alpha, \beta \in F; \forall x$

$\in V$

vector addition: $\alpha(x + y) = \alpha x + \alpha y$ $\forall \alpha \in F; \forall x, y \in V$

□ **Scalar identity** $1x = x$ $\forall x \in V$

Vector Space

Theorem (1)

Every vector space has a unique additive identity.

Every $v \in V$ has a unique additive inverse.



Vector Space

Example

Let V be the set of all real numbers with the operations $u \oplus v = u - v$ ($\oplus$ is an ordinary subtraction) and $\alpha \odot u = \alpha u$ ($\odot$ is an ordinary multiplication).

Is V a vector space? If it's not, which properties fail to hold?

Vector Space of functions

- Function addition and scalar multiplication

$$(f + g)(x) = f(x) + g(x) \quad \text{and} \quad (af)(x) = af(x)$$

Non-empty set X and any field F $\longrightarrow F^X = \{f: X \rightarrow F\}$

Example

- Set of all polynomials with real coefficients
- Set of all real-valued continuous function on $[0,1]$
- Set of all real-valued function that are differentiable on $[0,1]$

Vector Space of polynomials

$P_n(\mathbb{R})$: Polynomials with max degree (n)

- 
- Vector addition
 - Scalar multiplication
 - And other 8 properties!



Vector Space

Example

Which are vector spaces with $+$, $*$?

- ☐ Set $\mathbb{R}^n$ over $\mathbb{R}$
- ☐ Set $\mathbb{C}$ over $\mathbb{R}$
- ☐ Set $\mathbb{R}$ over $\mathbb{C}$
- ☐ Set $\mathbb{Z}$ over $\mathbb{R}$
- ☐ Set of all polynomials with coefficient from $\mathbb{R}$ over $\mathbb{R}$
- ☐ Set of all polynomials of degree at most n with coefficient from $\mathbb{R}$ over $\mathbb{R}$
- ☐ Matrix: $M_{m,n}(\mathbb{R})$ over $\mathbb{R}$
- ☐ Function: $f(x): x \rightarrow \mathbb{R}$ over $\mathbb{R}$

Conclusion

The operations on field F are:

- $+: F \times F \rightarrow F$
- $\times: F \times F \rightarrow F$



The operations on a vector space V over a field F are:

- $+: V \times V \rightarrow V$
- $\cdot: F \times V \rightarrow V$



04

Linear Combination

Linear Combinations

- The **linear combinations** of m vectors $a_1, \dots, a_m$, each with size n is:

$$\beta_1 a_1 + \dots + \beta_m a_m$$

where $\beta_1, \dots, \beta_m$ are scalars and called the **coefficients of the linear combination**

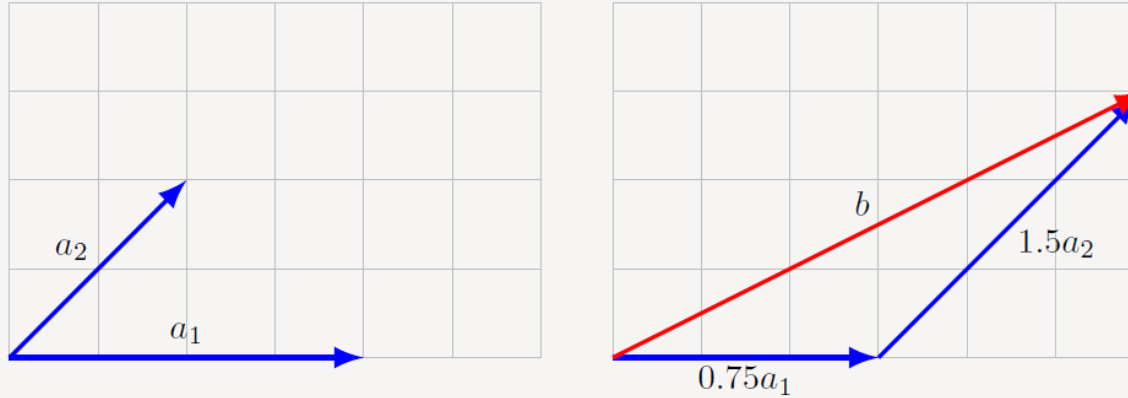
- **Coordinates**: We can write any n -vector b as a **linear combination of the standard unit vectors**, as:

$$b = b_1 e_1 + \dots + b_n e_n$$

- Example: What are the coefficients and combination for this vector?

$$\begin{bmatrix} -1 \\ 3 \\ 5 \end{bmatrix}$$

Linear Combinations



Left. Two 2-vectors a_1 and a_2 . Right. The linear combination $b = 0.75a_1 + 1.5a_2$

Special Linear Combinations

- ❑ Sum of vectors
- ❑ Average of vectors

05

Span – Linear Hull

Span or linear hull

Definition

If $v_1, v_2, v_3, \dots, v_p$ are in $\mathbb{R}^n$, then the set of all linear combinations of $v_1, v_2, \dots, v_p$ is denoted by $\text{Span}\{v_1, v_2, \dots, v_p\}$ and is called the **subset of $\mathbb{R}^n$ spanned (or generated) by $v_1, v_2, \dots, v_p$** .

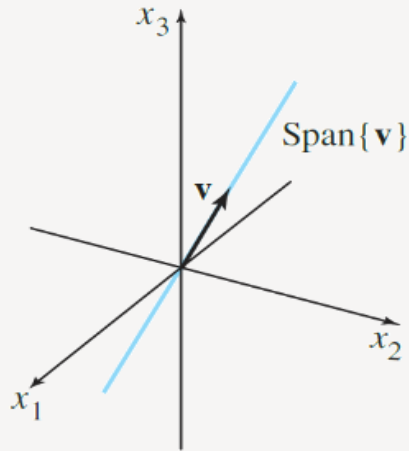
That is, $\text{Span}\{v_1, v_2, \dots, v_p\}$ is the collection of all vectors that can be written in the form:

$$c_1 v_1 + c_2 v_2 + \dots + c_p v_p$$

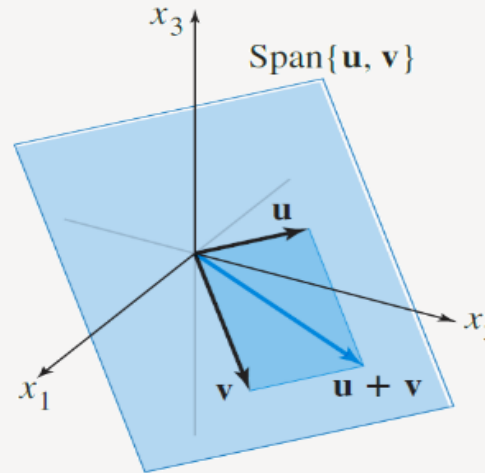
with $c_1, c_2, \dots, c_p$ being scalars.

Span Geometry

v and u are non-zero vectors in $\mathbb{R}^3$ where v is not a multiple of u

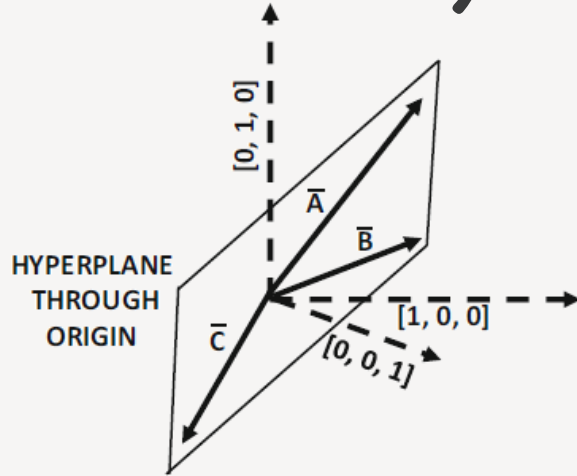


$\text{Span}\{v\}$ as a line through the origin.

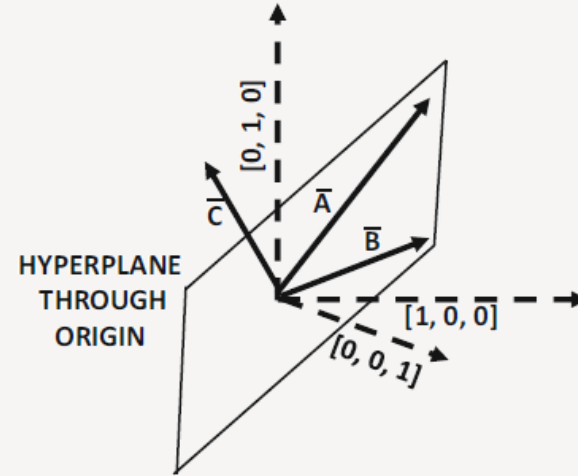


$\text{Span}\{u, v\}$ as a plane through the origin.

Span Geometry



(a) $\text{Span}(\{\vec{A}, \vec{B}\}) = \text{Span}(\{\vec{A}, \vec{B}, \vec{C}\})$
 $\text{Span}(\{\vec{A}, \vec{B}, \vec{C}\}) = \text{All vectors on hyperplane}$



(b) $\text{Span}(\{\vec{A}, \vec{B}\}) \neq \text{Span}(\{\vec{A}, \vec{B}, \vec{C}\})$
 $\text{Span}(\{\vec{A}, \vec{B}, \vec{C}\}) = \text{All vectors in } \mathbb{R}^3$

Figure 2.6: The span of a set of linearly dependent vectors has lower dimension than the number of vectors in the set

Span or linear hull

Example

- ❑ Is vector b in $\text{Span} \{v_1, v_2, \dots, v_p\}$
- ❑ Is vector v_3 in $\text{Span} \{v_1, v_2, \dots, v_p\}$
- ❑ Is vector 0 in $\text{Span} \{v_1, v_2, \dots, v_p\}$
- ❑ Span of polynomials: $\{(1+x), (1-x), x^2\}$?
- ❑ Is b in $\text{Span} \{a_1, a_2\}$?

$$a_1 = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}, a_2 = \begin{bmatrix} 5 \\ -13 \\ -3 \end{bmatrix}, b = \begin{bmatrix} -3 \\ 8 \\ 1 \end{bmatrix}$$

Resources

- ❑ Kenneth Hoffman and Ray A. Kunze. Linear Algebra. PHI Learning, 2004.
- ❑ David C. Lay, Steven R. Lay, and Judi J. McDonald. Linear Algebra and Its Applications. Pearson, 2016.
- ❑ Gilbert Strang. Introduction to Linear Algebra. Wellesley-Cambridge Press, 2016.

